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碩士論文計畫書
MASTER'S THESIS PROPOSAL

隱形帳簿：量化印尼平台經濟中的隱形楔子
The Invisible Ledger: Quantifying the Invisible Wedge in Indonesia's Platform Economy

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Abstract

The economic measurement of platform-based commerce rests on records that do not describe the same thing. Platform revenue, e-commerce statistics and payment data are commonly used as interchangeable indicators of digital activity, although each captures a different side of the boundary between the value a platform processes and the revenue it records. This paper measures that boundary for Indonesia, where I term it the invisible wedge, using issuer disclosures for 2020-2025. In FY2023, three documented platform cases processed transaction value 13.7 times their recognised revenue, a wedge of US\$40.07 billion or 2.9 percent of GDP, and the wedge stays above US\$20 billion when any single platform is excluded. Within platform series, revenue grew faster than transaction value in 10 of 12 annual transitions, with a median divergence of 42 percentage points. Official statistics place 98.5 percent of Indonesia's 2023-2024 e-commerce growth outside the marketplace channel that platform accounts record, while digital payment value grew up to eleven times as fast as e-commerce value. Each of these records misstates both the size of the digital economy and where it is growing.

1. Introduction

Southeast Asia's digital economy grew from approximately US\$100 billion in 2020 to US\$263 billion in 2024 (Google, Temasek and Bain 2020, 2024). Digital platform scale, however, is not the same thing as platform revenue. Platforms record transactions, determine participant payouts, and book their own revenue, while national statistics and payment systems record other parts of the same economic activity. These measures are often used interchangeably even though they do not measure the same thing. This paper argues that doing so can produce systematically wrong conclusions about the scale and growth of platform economies.

Suppose a platform processes 100 units of transaction value but recognizes only 10 as revenue. The remaining 90 are still recorded within the platform's system but do not constitute platform revenue; they may represent merchant receipts, driver payouts, inventory costs, taxes, and other pass-through payments. An analyst using the platform's accounts sees 10, while one measuring the commerce it processes sees 100. I call the difference the invisible wedge.

Indonesia provides a useful empirical setting because several records of the same activity can be compared. Platform companies disclose transaction and revenue information; BPS-Statistics Indonesia, the national statistical agency, reports e-commerce activity; and Bank Indonesia, the country's central bank, reports digital payments. In 2023, the platform comparison covers Shopee and Tokopedia, which together represented about 70 percent of estimated Indonesian marketplace transaction value, together with Grab, a major intermediary in delivery and other digital services. The three cases processed US\$43.23 billion of transaction value against US\$3.16 billion of recognized revenue, leaving a US\$40.07 billion invisible wedge.

The difference also changes over time. Across twelve year-to-year comparisons between 2020 and 2025, revenue grew faster than transaction value in ten, with a median divergence of approximately 42 percentage points; in some years the two measures moved in opposite directions. Platform revenue therefore understates the level of platform-mediated commerce while, in most observed years, overstating its growth.

The same problem appears beyond company accounts. BPS reports that national e-commerce value grew 17.1 percent between 2023 and 2024, but about 98.5 percent of that increase occurred outside the marketplace component; digital payment measures reported by Bank Indonesia grew substantially faster still. Indonesia's Minister of Finance Regulation No. 37 of 2025, which introduces seller reporting and withholding through designated marketplaces, provides a further test of how platform-held transaction records enter administrative reporting.

1.1 Research question and objectives

One question drives the proposal:

How much economic activity remains invisible when digital platforms are measured through their reported revenue?

Indonesia provides the main empirical setting. Three objectives follow: first, to measure the invisible wedge across available platform histories from 2020 to 2025; second, to explain large movements using disclosed changes in monetization, customer incentives, revenue recognition, and reporting scope; and third, to compare the platform evidence with national e-commerce and payment records to determine whether they produce the same account of economic scale and growth.

A separate comparison of eight platform businesses outside Indonesia tests whether the same transaction-revenue separation appears under other business models.

1.2 Contributions

First, the study develops the invisible wedge from a one-year estimate into a measure that can be followed over time. From 2020 to 2025, revenue grows faster than transaction value in ten of twelve annual comparisons, showing that platform revenue is not a stable measure of either the level or the growth of platform-mediated commerce.

Second, I examine why the wedge moves using accounting components disclosed by the platforms themselves. Between 2022 and 2023, Tokopedia's transaction value fell 8.9 percent while selected third-party net revenue rose 53.2 percent. About 60.6 percent of the arithmetic increase in net revenue is associated with lower customer incentives, with the remainder associated with higher gross revenue.

Third, I extend the comparison beyond company accounts. National e-commerce statistics show that almost all of Indonesia's 2023–2024 growth occurred outside the marketplace component, while digital payment measures grew substantially faster than e-commerce value. The records therefore produce different conclusions about the scale, growth, and location of digital activity.

Fourth, I test whether the main findings survive alternative assumptions, sample rules, and external platform evidence rather than depending on one preferred construction.

1.3 Positioning the contribution

The contribution is a finance measurement problem before it is a tax-policy one. Platform accounts make the intermediation boundary observable, but that boundary affects the conclusions investors, researchers, statistical agencies, and policymakers draw about scale, growth, and monetization. De Franco, Kothari and Verdi (2011) show why comparable accounting bases matter, while Berg et al. (2020) show that intermediary-held digital traces contain information conventional records can miss. This paper applies those ideas to platform economies and shows that the measure chosen to describe digital activity can change the economic conclusion.

2. Literature Review and Research Gap

Digital platforms sit between economic activity and the records used to measure it. They facilitate transactions between buyers and sellers, recognize only part of that activity as their own revenue, and generate records that can also appear in payment, statistical, and administrative systems. Existing research explains each part of this process from a different angle. Taken together, it provides the foundation for asking how much activity remains invisible when platforms are measured through reported revenue.

2.1 Platform Economics and Revenue Recognition

Platform economics explains why the value of commerce processed through a platform can be much larger than the revenue the platform records. Digital platforms often connect consumers with merchants or other service providers and earn commissions, fees, advertising, logistics, or other intermediation revenue rather than the full value of every transaction they facilitate (Rochet and Tirole 2003; Caillaud and Jullien 2003; Parker and Van Alstyne 2005; Armstrong 2006; Hagiu and Wright 2015; Evans and Schmalensee 2016).

Accounting can widen or narrow this difference. Under IFRS 15, whether a company acts as a principal or an agent affects whether it recognizes the full customer payment or only the amount it earns from arranging the transaction (International Accounting Standards Board 2014). Customer incentives, monetization, service mix, acquisitions, and changes in business scope can also move reported revenue without an equivalent change in transaction activity. De Franco, Kothari and Verdi (2011) show why comparable accounting bases matter when financial information is interpreted. The literature therefore explains why a transaction–revenue gap can exist and move over time, but not how economically large that gap becomes in practice.

2.2 Digital Records and Third-Party Information

Activity outside platform revenue is not necessarily unrecorded. Merchants and service providers can be small, self-employed, or weakly represented in conventional financial records while their transactions still leave detailed records inside a digital platform. This differs from the usual shadow-economy problem, where part of the difficulty is that economic activity leaves few reliable formal records (La Porta and Shleifer 2014; Ulyssea 2018; Medina and Schneider 2019). Digital records can contain useful economic information that conventional records miss (Berg et al. 2020), and platform-mediated activity can create additional financial and administrative traces (Barrios, Hochberg and Yi 2022; Denes, Lagaras and Tsoutsoura 2025).

These records also matter once they become available to government. Public-finance research shows that compliance and enforcement change when information about income or transactions is independently reported by third parties (Kleven et al. 2011; Pomeranz 2015; Naritomi 2019; Kleven, Kreiner and Saez 2016; Slemrod 2019). The OECD Model Rules and the European Union's DAC7 regime apply this principle to digital platforms. Indonesia's Minister of Finance Regulation No. 37 of 2025 follows the same logic through seller identification, transaction-linked reporting, and marketplace withholding. The relevance here is informational: transaction records outside platform revenue may still exist and become usable elsewhere.

2.3 Measuring the Digital Economy

Official measurement frameworks already distinguish between the commerce facilitated through a platform and the service provided by the platform itself. National-accounting guidance separates the buyer-seller transaction from the platform's intermediation service rather than treating the full transaction value as platform output (Ahmad and Schreyer 2016; International Monetary Fund 2018; OECD 2023; United Nations et al. 2025).

In practice, however, gross transaction value, platform revenue, e-commerce statistics, and payment data are all used to describe digital economic activity. They measure different things. A difference that is harmless when the measures move together becomes important when they do not, because the choice of measure can change conclusions about economic scale and growth. This thesis therefore compares the records rather than assuming that one can substitute for another.

2.4 Research Gap

Existing research explains the pieces of the problem separately. Platform economics explains why transaction value can exceed revenue; accounting explains why the gap can move; research on digital records explains why activity outside revenue can still be observable; and official measurement frameworks explain why these quantities should not be treated as equivalent.

What remains less clear is how large these differences become in practice, whether they persist over time, and whether choosing one measure instead of another changes the economic conclusion. This thesis addresses that gap by measuring the invisible wedge over time, examining disclosed reasons for major movements, comparing platform accounts with national e-commerce and payment records, and using external platform evidence to test whether the same measurement problem appears beyond the main Indonesian setting.

3. Theoretical Framework and Key Variables

Let V denote transaction value and R platform-recognized revenue for a matched platform scope and period. The core measures are:

$$W = V - R \quad E = (V - R) / R \quad D = g(V) - g(R)$$

W is the absolute invisible wedge; E , the Ecosystem Ratio, is constructed for this study and measures the wedge per unit of platform revenue; and D is the annual growth divergence between transaction value and revenue. A negative D means platform revenue is growing faster than the commerce behind it.

Gross transaction value (GTV) is the proposal's common label for the value processed through a platform. Issuer terminology differs: Sea Limited and Blibli use GMV, Grab uses GMV for deliveries and TPV for financial services, and GoTo uses GTV. The source labels and business perimeters are retained in the empirical files, and the main longitudinal comparisons are made within named series.

3.1 Hypotheses

The framework yields four hypotheses, set out in Table 1 with the evidence already assembled for each. H4 is the central claim of the thesis, and H1 to H3 establish the mechanisms behind it. Whether platform records become administratively usable once PMK 37/2025 takes effect on 1 November 2026 depends on post-implementation evidence; Section 5.3 addresses it through the design of the regime.

Table 1. Hypotheses and current evidence.

Hypothesis	Status in the current evidence
H1 Platform revenue is not a stable proxy for platform commerce: the ratio of transaction value to revenue moves materially within the same platform over time.	Supported. Revenue outgrows transaction value in 10 of 12 within-series transitions; median divergence 42.02 pp. Survives log growth (36.64), removal of the largest transition (29.34 pp) and exclusion of any single series.
H2 Indonesian e-commerce growth is driven by the entry of new businesses rather than by higher value per business.	Supported. The business-count term accounts for 90.29% of the 2023-2024 increase in e-commerce value, and 70.95% under the conflicting BPS business count.
H3 Marketplace participation is associated with financial recordkeeping.	Supported in BPS business-level evidence. The within-province change is weaker ($r = 0.309$, $p = 0.066$) and is tested further in the thesis.
H4 Platform revenue, marketplace statistics and payment data are not interchangeable measures of digital commerce, and substituting one for another misstates its scale and location.	Supported. FY2023 transaction value is 13.7 times revenue; 98.46% of 2023-2024 e-commerce growth falls outside the marketplace channel; payment value grew 1.8-11 times as fast as e-commerce value.

4. Data and Methodology

4.1 Admission rules and evidence tiers

A platform-period is eligible only when transaction value and revenue cover the same period; geography and business scope can be evaluated; units and definitions are known; derived inputs trace to source; and structural breaks or revised reporting bases are flagged. Repeated publication vintages are not counted as independent observations. Final admission to the main sample is decided at the proposal examination (Section 7).

Table 2. Evidence tiers for Indonesian platform observations. Tiers are reported separately according to how directly each series maps to Indonesia.

Evidence tier	Series	Levels	Transitions
Direct Indonesia-aligned	Tokopedia e-commerce	2	1
Direct, scope-pending	Blibli 3P; Bukalapak Group	9	7
Conditional reconstruction	Grab; Shopee Indonesia	6	4
All tiers (diagnostic)	Five series	17	12

Table 2 shows the trade-off the design manages: only two of the seventeen levels come from a directly Indonesia-aligned segment, while the larger tiers carry more observations under wider scope. Tokopedia is the strongest Indonesia-aligned series; Blibli includes online travel; Bukalapak reports at Group level with overseas operations; and Grab and Shopee each require a derived or externally estimated country component. Tokopedia FY2021 and Bukalapak FY2024 are excluded because their transaction and revenue periods do not match.

Three counts are intentionally different. The all-tier inventory contains 17 retained levels; the executed direct tiers contain 11 levels (2 Indonesia-aligned and 9 scope-pending); and the broader direct-candidate sensitivity contains 13 periods, including a Blibli FY2020 prospectus observation, yielding 9 annual transitions. These counts answer different construction questions and are reported separately.

4.2 Comparability strategy

The main longitudinal comparisons are made within named series. A within-series transition holds the issuer, business perimeter and disclosure convention as constant as the filings allow, so a change in the ratio of transaction value to revenue reflects a change in that platform's economics and accounting.

Cross-platform levels serve a different purpose. The FY2023 cross-section establishes scale, and the 48 external issuer-years test whether the boundary appears outside Indonesia. Keeping those roles distinct is what allows heterogeneous disclosures to be used together.

4.3 Conditional country evidence

Grab discloses Indonesia revenue but not Indonesia transaction value. Its conditional country transaction value is therefore derived as:

$$\text{Indonesia GTV} = \text{Indonesia revenue} \times \text{Group GTV} / \text{Group revenue}$$

The construction applies the Group monetisation rate to Indonesia. Shopee's Indonesian transaction value comes from Momentum Works' recurring Southeast Asian e-commerce estimates, since Sea Limited discloses no country figure, and is labelled as third-party wherever it is used.

5. Preliminary Evidence and Feasibility

5.1 FY2023 scale

Table 3. Indonesia-focused FY2023 cross-section, US\$ billions. *V* is transaction value, *R* platform revenue, *W* the wedge and *E* the Ecosystem Ratio; the final row sums the three cases, and *E* is computed from unrounded inputs.

Case	V	R	W	E	Evidence class
Grab Indonesia	5.381	0.605	4.776	7.895x	Derived V
Tokopedia	16.331	0.405	15.926	39.296x	Direct pair
Shopee Indonesia	21.520	2.152	19.368	9.000x	Derived V and R
Selected platforms	43.233	3.162	40.070	12.671x	Sum of three cases

Table 3 reports the FY2023 cross-section. Across the three cases, transaction value is 13.7 times recognised revenue. Tokopedia, the one direct pair, contributes 39.7 percent of the wedge, Shopee 48.3 percent and Grab 11.9 percent. Shopee's Indonesian revenue is estimated by applying Sea Limited's disclosed Group monetisation rate of 10.0 percent to an external market estimate, which fixes its Ecosystem Ratio at 9.000; varying that rate between 9 and 11 percent moves the combined wedge only between US\$40.29 billion and US\$39.86 billion, because a higher assumed rate raises estimated revenue and lowers the residual almost equally. Excluding any single platform leaves a wedge of between US\$20.70 billion and US\$35.29 billion, and one-at-a-time variation of every derived input keeps it between US\$37.65 billion and US\$42.49 billion. Tokopedia alone, which rests on no reconstruction, carries a wedge of US\$15.93 billion.

5.2 Longitudinal movement and mechanism

Table 4. Within-series annual divergence under two admission rules. Counts differ because the admission rules differ.

	All-tier inventory	Direct-candidate sensitivity
Transitions	12	9
Revenue grows faster	10	6
Transaction grows faster	2	3
Opposite-sign transitions	2	3
Median absolute divergence	42.02 pp	42.94 pp

Table 4 compares the two admission rules, and both lead to the same conclusion. Under the all-tier inventory, revenue grew faster than transaction value in 10 of 12 annual transitions, with a median absolute divergence of 42.02 percentage points; under the direct-candidate rule, revenue grew faster in 6 of 9 and the median is 42.94 points. Platform revenue growth is therefore a poor guide to the growth of the commerce behind it, and in most years it overstates that growth. Under the direct-candidate rule the result survives the obvious tests: measured in log changes the median gap is 36.64 points, removing the single largest transition leaves 29.34 points, and every construction that excludes one platform series retains at least one sign reversal. Tokopedia FY2022-FY2023 shows the mechanism: transaction value fell 8.90 percent while third-party net segment revenue rose 53.20 percent, and 60.56 percent of that revenue gain came from lower customer incentives, with 39.44 percent from higher gross revenue.

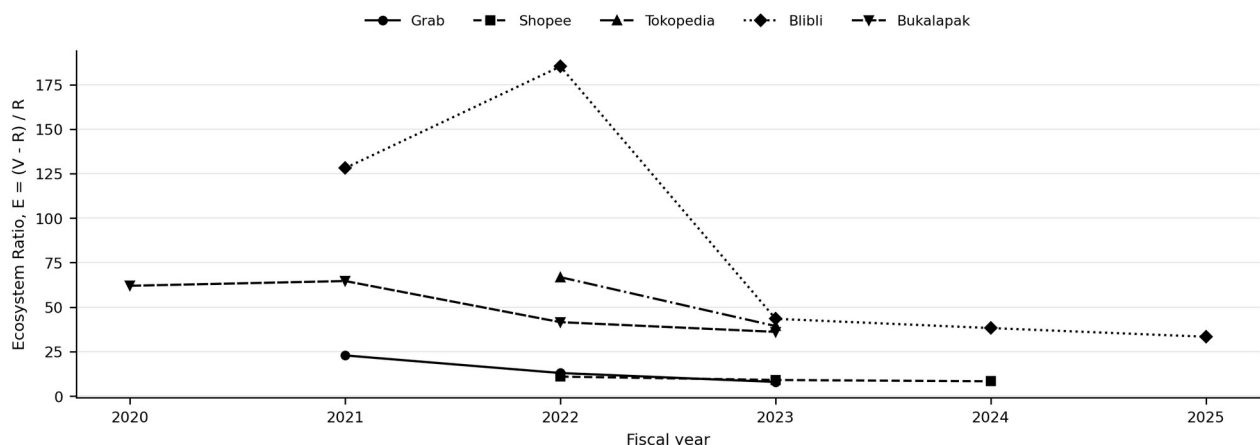


Figure 1. Ecosystem Ratio across the extended candidate series, FY2020-FY2025. Each line follows one disclosed or reconstructed series.

Figure 1 shows the pattern behind Table 4. The wedge is present in every series in every year, at ratios from about 8 to over 180, and in every series it is lower at the end of the period than at the start, as platforms have raised their monetisation. Measured through platform revenue, these businesses appear to be growing faster than the commerce they carry, which is the distortion H1 describes.

5.3 Wider Indonesian activity and reporting

BPS reports national e-commerce transaction value rising 17.08 percent from 2023 to 2024. The marketplace component, the channel that platform accounts describe, rose 1.45 percent while the non-marketplace component rose 20.57 percent, so 98.46 percent of the increase took place outside the marketplace channel. The marketplace share of e-commerce value fell from 18.2 to 15.8 percent in a single year, so measurement anchored on marketplace platforms misses almost all of the recent growth. Growth in the number of e-commerce businesses accounts for 90.29 percent of the national increase, with value per business contributing the remainder (H2).

PMK 37/2025 is the institutional response. The verified implementation sequence records marketplace designation on 1 July 2026, collection effective 1 August, postponement through 31 October, and scheduled implementation on 1 November 2026. A platform record becomes usable

for tax administration only once a seller identity is attached, the record is transmitted to the DJP and it is matched to a taxpayer. The regime designates marketplace operators, and the marketplace channel carried 15.8 percent of 2024 e-commerce value and 1.5 percent of its 2023-2024 growth. However well it is implemented, its coverage is bounded by the channel it targets.

Outside Indonesia, 48 matched issuer-years across eight platform businesses, including eBay, Etsy, Shopify, Mercado Libre and Sea, show the same boundary under other business models. Eleven of 40 annual transitions fail a clean-scope test because of acquisitions or perimeter changes, which indicates that the comparability problem the evidence tiers address is a feature of platform disclosure generally.

5.4 Payment traces and administrative records

Bank Indonesia's payment statistics provide a third record of the same activity. Between 2023 and 2024, BPS e-commerce value grew 17.08 percent, while electronic-money shopping value grew 30.47 percent, mobile-banking payment and purchase value 82.84 percent and QRIS transaction value 186.98 percent. Payments and sales are different objects, and the payment series grew between 1.8 and 11 times as fast as the commerce they help settle, so a digital-economy indicator built on payment value would substantially overstate commerce growth over this period.

5.5 What the preliminary evidence establishes

The preliminary evidence supports the central claim at each step. Platform revenue understates the level of platform commerce by a factor of 13.7 and, in most years, overstates its growth. Marketplace statistics miss 98.5 percent of the recent expansion. Payment data grow up to eleven times as fast as the commerce behind them. And the one disclosed decomposition traces the revenue-transaction gap to identifiable commercial choices. The thesis will extend and test these results under the sample boundary approved at examination.

6. Why the Problem Matters

The consequences follow directly. An investor valuing platforms on revenue growth, a statistical agency sizing e-commerce from marketplace data and a policymaker reading payment growth as commerce growth are each working from a record that departs sharply from the activity they intend to measure, and in a different direction. The same transformation looks like rapidly monetising platforms, a stagnant marketplace sector or an explosion of digital commerce depending on which record is used. Indonesia's new reporting regime inherits the problem, because it is built around the marketplace channel. The thesis identifies proxy substitution between these records as the underlying error, and measures its size.

7. Robustness and Proposal Decisions

Table 5. Claims, evidence and robustness tests.

Claim	Evidence and robustness test
Platform revenue understates platform commerce by more than an order of magnitude.	FY2023 transaction value is 13.7 times revenue (wedge US\$40.07bn). Holds across Shopee monetisation of 9-11% (US\$39.86-40.29bn), leave-one-platform-out (at least US\$20.70bn), one-at-a-time input variation (US\$37.65-42.49bn), and on the Tokopedia direct pair alone (US\$15.93bn).
Platform revenue growth misstates commerce growth.	Revenue outgrows transaction value in 10 of 12 transitions; median divergence 42.02 pp. Holds in log changes (36.64), without the largest transition (29.34 pp), under the direct-candidate rule (42.94 pp) and in every leave-one-series-out construction.
Marketplace statistics miss where e-commerce grows.	98.46% of 2023-2024 growth in e-commerce value falls outside the marketplace component; marketplace share 18.2% to 15.8%. Holds on both the published amount and the rounded-share reconstruction (98.49%).
Payment data overstate commerce growth.	Electronic-money, mobile-banking and QRIS payment value grew 1.8 to 11 times as fast as e-commerce value, 2023-2024.
Questions the thesis tests further.	Causal effects of incentive choices; business-level validation of the BPS recordkeeping association (province $r = 0.309$, $p = 0.066$); operational performance of PMK 37/2025 after 1 November 2026.

Table 5 sets each claim against its evidence and the robustness tests it survives. The sample is small and the evidence descriptive, so the thesis reports tests rather than population estimates. What it establishes is that the standard records of digital commerce diverge by magnitudes too large for measurement or policy to ignore.

Two sample decisions are put to the committee: whether Blibli 3P Retail and Bukalapak Group enter the final longitudinal core, and whether Grab and Shopee remain sensitivity evidence. The central claim holds under either outcome: the divergence result holds under both admission rules in Table 4, and the level result holds on the Tokopedia pair alone.

8. Next Step

After the proposal examination the approved sample boundary will be frozen and the requested checks completed during October-November 2026, with the full manuscript rebuilt around that boundary before the final defence. Robustness, ASEAN/payment extensions, detailed source reconciliation, and the complete verification ledger remain available in the repository and advisor data package. Table 6 sets out that schedule; the empirical backend is already assembled, so what remains is sample admission, requested analysis, and manuscript construction.

Table 6. Planned work to the final defence.

Period	Planned work
September 2026	Advisor review of the empirical package; proposal oral examination; freeze the sample-admission rule and the checks the committee requests.
October-November 2026	Complete requested reconciliations and source concordance; extend sensitivity to the approved tier boundary; freeze thesis tables and figures.
Before the final defence	Rebuild the manuscript under the approved sample; retain supporting payment, ASEAN and global modules only where they advance the final argument.

The proposal establishes that the records used to measure Indonesia's platform economy diverge far more than their interchangeable use assumes. Platform revenue understates the level of platform commerce by more than an order of magnitude and misstates its growth, marketplace statistics miss almost all of the recent expansion, and payment data outrun commerce by up to eleven times. What remains is to fix the sample boundary, complete the checks the committee requires and build the thesis on that decision; the data, the evidence tiers and the reproducible backend are already in place.

Appendix A - data sources by evidence class

Table A1. Source lineage by evidence class.

Tier	Series	Source of V	Source of R
Direct Indonesia-aligned	Tokopedia e-commerce	GoTo annual report, e-commerce segment metrics	GoTo annual report, segment note
Direct, scope-pending	Blibli 3P Retail	Global Digital Niaga prospectus and results releases	Same issuer filings
Direct, scope-pending	Bukalapak Group	Bukalapak annual and sustainability reports	Same issuer filings
Conditional reconstruction	Grab Indonesia	Derived from disclosed Indonesia revenue and Group monetization rate (Form 20-F)	Grab Form 20-F
Conditional reconstruction	Shopee Indonesia	Momentum Works Southeast Asia e-commerce estimate (third-party)	Derived from Sea Limited Form 20-F
National statistics	BPS-Statistics Indonesia	E-Commerce Statistics 2023, 2024; BPS directorate presentation	-
Regulatory	PMK 37/2025; DJP	Ministry of Finance of the Republic of Indonesia	-

Every figure used in this proposal traces to a named file and locator in the accompanying data package.

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